

Quadratic Parameters, Tangency, and Constrained Extrema

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§1.1 A scattered algebraic note

Elementary-looking algebraic problems often hide a parameter, a tangency condition, or an extremum under constraint. The problems below are organized around those three mechanisms.

§1.2 A quadratic with parameters

Let

$$f(x) = ax^2 + bx + c, \quad a < b < c,$$

and suppose the graph passes through $A = (1, 0)$ and $B = (m, -a)$. The condition $f(1) = 0$ gives

$$a + b + c = 0.$$

The condition $f(m) = -a$ is equivalent to saying that $f(x) + a$ has $x = m$ as a root:

$$am^2 + bm + c + a = 0.$$

Using $c = -a - b$, this becomes

$$am^2 + bm - b = 0, \quad am^2 = -b(m - 1).$$

The axis of symmetry is

$$x_0 = -\frac{b}{2a}.$$

Thus the parameter m and the axis x_0 are tied by

$$x_0 = \frac{m^2}{2m - 2}.$$

Equivalently,

$$m^2 = (2m - 2)x_0, \quad m^2 - 2x_0m + 2x_0 = 0.$$

Solving for m gives

$$m = x_0 \pm \sqrt{x_0^2 - 2x_0}.$$

Use this relation to examine endpoints and stationary points of m as a function of x_0 . The extremal parameter occurs at

$$x_0 = 0 \quad \text{or} \quad x_0 = -\frac{1}{2},$$

and the relevant value is

$$m = \frac{\sqrt{5} - 1}{2}.$$

This is a small example of a recurring trick: convert the algebraic parameter into a geometric parameter, usually the axis, vertex, or tangency point.

§1.3 A parabola and a hyperbola

The second problem considers the parabola

$$y = x^2 - 2ax + a^2 - 4 = (x - a)^2 - 4$$

and the rectangular hyperbola

$$y = \frac{k}{x}, \quad k > 0,$$

in the first quadrant. The intersection equation is

$$x^2 - 2ax + a^2 - 4 = \frac{k}{x}, \quad x > 0,$$

or

$$x^3 - 2ax^2 + (a^2 - 4)x - k = 0.$$

Questions about “only one intersection” are therefore tangency or root-counting questions for this cubic, but the geometry is usually cleaner: one restricts to the right side of the parabola, because the branch k/x lies in the first quadrant.

Remark 1.3.1 — The hidden principle is that a graph-intersection problem may become either a polynomial root-counting problem or a tangency problem. The latter is often more geometric and less computational.

§1.4 A constrained extremum

The third problem asks for extrema of

$$x + y$$

under the constraint

$$x^2 + y^3 = 2.$$

Using Lagrange multipliers for

$$F(x, y, \lambda) = x + y + \lambda(x^2 + y^3 - 2),$$

we get

$$1 + 2\lambda x = 0, \quad 1 + 3\lambda y^2 = 0, \quad x^2 + y^3 = 2.$$

Eliminating λ gives

$$2x = 3y^2.$$

Hence

$$x = \frac{3}{2}y^2.$$

Substitution into the constraint gives

$$\frac{9}{4}y^4 + y^3 = 2,$$

so the stationary points come from the quartic

$$9y^4 + 4y^3 - 8 = 0.$$

Numerical solution gives approximately

$$A = (1.15195, 2.02829), \quad B = (1.82947, 0.72509),$$

for the function

$$h(x) = x + (2 - x^2)^{1/3}.$$

These are the local maximum and local minimum along the real branch.

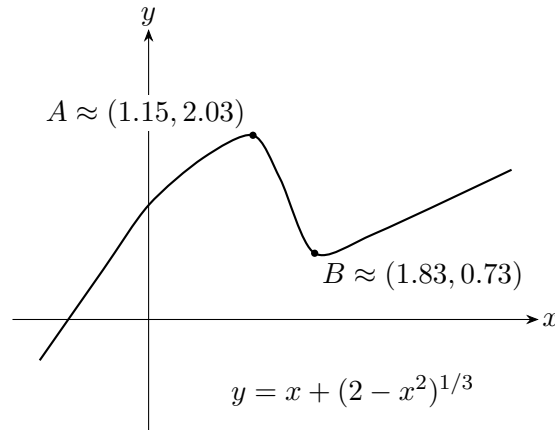


Figure 1.1: Reconstruction of the graph used for the constrained extremum problem.

§1.5 Tangency as a double-root condition

A parameter problem involving a quadratic often asks when a line is tangent to a parabola or when an equation has exactly one solution. Algebraically, this is a discriminant condition:

$$\Delta = b^2 - 4ac = 0.$$

Geometrically, the line touches the parabola at one point. The same idea appears in constrained extrema: a level curve just touches a constraint curve at an optimum.

§1.6 Completing the square

For a quadratic

$$ax^2 + bx + c, \quad a \neq 0,$$

completing the square gives

$$a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a}.$$

This form reveals the vertex, minimum or maximum value, and symmetry axis. It is usually more informative than the expanded form.

§1.7 A wider frame

The three examples are connected by the same idea: reduce a complicated statement to a structural condition. The axis of a quadratic, the tangency of two graphs, and the Lagrange multiplier equations are all ways of saying that an extremum occurs when a first-order freedom disappears.

§1.8 Tangency as multiplicity

A line is tangent to a curve when the intersection equation has a repeated root. If a family is written as

$$F(x, t) = 0,$$

then a tangency or transition value often satisfies

$$F(x, t) = 0, \quad \frac{\partial F}{\partial x}(x, t) = 0.$$

For a quadratic, this is the discriminant condition $\Delta = 0$. The same idea persists for higher-degree equations: a repeated root is detected by a common factor of F and F_x .

§1.9 Constrained extrema and normal directions

For a constraint $g(x, y) = 0$, the Lagrange multiplier condition

$$\nabla f = \lambda \nabla g$$

says that at an interior constrained extremum, the gradient of f has no component tangent to the constraint curve. Equivalently, the differential df vanishes on the tangent space of the constraint.

This is the coordinate-free meaning of the method: extrema occur where the objective's first-order change is normal to every allowed first-order motion.

§1.10 Degeneracy and second-order data

After first derivatives vanish, the next question is second-order behavior. For an unconstrained two-variable function, the Hessian quadratic form determines the local model:

$$f(a + h) = f(a) + \frac{1}{2}h^T H_f(a)h + o(\|h\|^2).$$

For constrained problems, the Hessian must be restricted to tangent directions. Degenerate cases occur when this quadratic form has zero directions, forcing higher-order analysis.

§1.11 Multiplicity, normality, and local geometry

Quadratic parameter problems, tangency, and constrained extrema are one theme: detect where the number or type of solutions changes. Algebra sees this as multiplicity or discriminant zero; geometry sees it as tangency; optimization sees it as vanishing first variation plus second-order testing.